WI4014TU – Numerical Analysis for PDE's

Lecture 12
Numerical solution of the wave equation

Scalar wave equation

$$
\begin{aligned}
&\frac{1}{c^2}\frac{\partial^2 u}{\partial t^2} - \Delta u = f, \quad \boldsymbol{x} \in \Omega, \;\; 0 < t \leq T \\
&u(\boldsymbol{x}, t) = g(\boldsymbol{x}, t), \quad \boldsymbol{x} \in \partial\Omega, \;\; 0 < t \leq T \\
&u(\boldsymbol{x}, 0) = u_0(\boldsymbol{x}), \quad \boldsymbol{x} \in \Omega, \;\; t = 0 \\
&\left.\frac{\partial u(\boldsymbol{x}, t)}{\partial t}\right|_{t=0} = v_0(\boldsymbol{x}), \quad \boldsymbol{x} \in \Omega, \;\; t = 0
\end{aligned}
$$

Method of lines for wave equation

With FDM/FVM spatial discretization

Discretization in space: $-\Delta u \to A\boldsymbol{u}$, $\boldsymbol{u}(t) \in \mathbb{R}^n$, $A \in \mathbb{R}^{n\times n}$

$$\boldsymbol{u}'' = -c^2 A\boldsymbol{u}, \quad \boldsymbol{u}(0) = \boldsymbol{u}^0, \;\; \boldsymbol{u}'(0) = \boldsymbol{w}^0$$

Assume $A = V\Lambda V^{-1}$ exists. Then, $\boldsymbol{u}(t) = V\boldsymbol{b}(t)$, and $\boldsymbol{b}(t) = V^{-1}\boldsymbol{u}(t)$.

$$\begin{aligned}
\boldsymbol{b}'' &= -c^2\Lambda\boldsymbol{b}, \quad \boldsymbol{b}(0) = V^{-1}\boldsymbol{u}^0, \;\; \boldsymbol{b}'(0) = V^{-1}\boldsymbol{w}^0, \\
b_j''(t) &= -c^2\lambda_j b_j(t), \quad j = 1, \dots, n \\
b_j(t) &= a_j e^{ic\sqrt{\lambda_j}\, t} + d_j e^{-ic\sqrt{\lambda_j}\, t} \\
\boldsymbol{b}(t) &= e^{ic\Lambda^{1/2}t}\boldsymbol{a} + e^{-ic\Lambda^{1/2}t}\boldsymbol{d}, \quad \text{use initial conditions} \\
\boldsymbol{b}(t) &= \frac{1}{2}\left(e^{ic\Lambda^{1/2}t} + e^{-ic\Lambda^{1/2}t}\right)\boldsymbol{b}(0) \\
&\quad + \frac{1}{2i}\left(e^{ic\Lambda^{1/2}t} - e^{-ic\Lambda^{1/2}t}\right)\frac{1}{c}\Lambda^{-1/2}\boldsymbol{b}'(0)
\end{aligned}$$

Method of lines solution

$$
\begin{aligned}
\boldsymbol{u}'' &= -c^2 A\boldsymbol{u}, \quad \boldsymbol{u}(0) = \boldsymbol{u}^0, \;\; \boldsymbol{u}'(0) = \boldsymbol{w}^0, \\
\boldsymbol{u}(t) &= V\boldsymbol{b}(t), \quad \boldsymbol{u}(0) = \boldsymbol{u}^0, \;\; \boldsymbol{u}'(0) = \boldsymbol{w}^0 \\
\boldsymbol{u}(t) &= V\left[\cos\left(c\Lambda^{1/2}t\right)\boldsymbol{b}(0) + \sin\left(c\Lambda^{1/2}t\right)\frac{1}{c}\Lambda^{-1/2}\boldsymbol{b}'(0)\right] \\
&= \sum_{j=1}^{n}\left[b_j(0)\cos\left(ct\sqrt{\lambda_j}\right) + b_j'(0)\frac{1}{c\sqrt{\lambda_j}}\sin\left(ct\sqrt{\lambda_j}\right)\right]\boldsymbol{v}_j
\end{aligned}
$$

Conclusion: oscillating behavior, evolution coefficients neither grow nor decay with time (with the diffusion equation we had exponential decay).

Numerical time "integration" of the wave equation

Central-difference time discretization

$$\frac{\partial^2 u}{\partial t^2} = \frac{u^{k+1}(\boldsymbol{x}) - 2u^k(\boldsymbol{x}) + u^{k-1}(\boldsymbol{x})}{h_t^2} + \mathcal{O}(h_t^2)$$

Explicit time-stepping scheme:

$$\frac{u^{k+1}(\boldsymbol{x}) - 2u^k(\boldsymbol{x}) + u^{k-1}(\boldsymbol{x})}{c^2 h_t^2} = \Delta u^k(\boldsymbol{x}) + f^k(\boldsymbol{x})$$
$$u^{k+1}(\boldsymbol{x}) = 2u^k(\boldsymbol{x}) - u^{k-1}(\boldsymbol{x}) + (ch_t)^2 \, \Delta u^k(\boldsymbol{x}) + (ch_t)^2 \, f^k(\boldsymbol{x})$$

Needs both $u^k(\boldsymbol{x})$ and $u^{k-1}(\boldsymbol{x})$ to run

Approximation of initial conditions

To start the iterations we need $u^1(\boldsymbol{x})$ and $u^0(\boldsymbol{x})$. The initial conditions are

$$u(\boldsymbol{x}, 0) = u_0(\boldsymbol{x}), \quad \left.\frac{\partial u(\boldsymbol{x}, t)}{\partial t}\right|_{t=0} = v_0(\boldsymbol{x}), \quad \boldsymbol{x} \in \Omega$$

Hence, $u^0(\boldsymbol{x}) = u_0(\boldsymbol{x})$. To approximate $u^1(\boldsymbol{x})$ consider the Taylor expansion

$$u^1(\boldsymbol{x}) = u^0(\boldsymbol{x}) + \left.\frac{\partial u(\boldsymbol{x}, t)}{\partial t}\right|_{t=0} h_t + \frac{1}{2} \left.\frac{\partial^2 u(\boldsymbol{x}, t)}{\partial t^2}\right|_{t=0} h_t^2 + \mathcal{O}(h_t^3)$$

Since one-sided FD approximation

$$\frac{u^1(\boldsymbol{x}) - u^0(\boldsymbol{x})}{h_t} = \left.\frac{\partial u(\boldsymbol{x}, t)}{\partial t}\right|_{t=0} + \mathcal{O}(h_t)$$

gives only $\mathcal{O}(h_t)$ approximation of the initial condition, we need to retain one more term in the approximation of $u^1(\boldsymbol{x})$

$\mathcal{O}(h_t^2)$ approximation of initial conditions

$$
\begin{aligned}
u^1(\boldsymbol{x}) &= u^0(\boldsymbol{x}) + \left.\frac{\partial u(\boldsymbol{x},t)}{\partial t}\right|_{t=0} h_t + \frac{1}{2}\left.\frac{\partial^2 u(\boldsymbol{x},t)}{\partial t^2}\right|_{t=0} h_t^2 + \mathcal{O}(h_t^3) \\
&= u_0(\boldsymbol{x}) + h_t v_0(\boldsymbol{x}) + \frac{(ch_t)^2}{2}\left[\Delta u_0(\boldsymbol{x}) + f^0(\boldsymbol{x})\right] + \mathcal{O}(h_t^3)
\end{aligned}
$$

where we have used the wave equation at $t = 0$:

$$
\frac{1}{c^2}\left.\frac{\partial^2 u}{\partial t^2}\right|_{t=0} = \Delta u(\boldsymbol{x},0) + f(\boldsymbol{x},0)
$$

Hence, given $u_0(\boldsymbol{x})$, $v_0(\boldsymbol{x})$, and $f^k(\boldsymbol{x})$, the time-stepping iterations are:

$$
\begin{aligned}
u^0(\boldsymbol{x}) &= u_0(\boldsymbol{x}) \\
u^1(\boldsymbol{x}) &= u_0(\boldsymbol{x}) + h_t v_0(\boldsymbol{x}) + \frac{(ch_t)^2}{2}\left[\Delta u_0(\boldsymbol{x}) + f^0(\boldsymbol{x})\right] \\
u^{k+1}(\boldsymbol{x}) &= 2u^k(\boldsymbol{x}) - u^{k-1}(\boldsymbol{x}) + (ch_t)^2\left[\Delta u^k(\boldsymbol{x}) + f^k(\boldsymbol{x})\right]
\end{aligned}
$$

Time-stepping with FDM/FVM Laplacian

After discretization in space $-\Delta u^k(\boldsymbol{x}) \to A\boldsymbol{u}^k$:

$$\boldsymbol{u}^0, \quad \boldsymbol{w}^0 \quad - \quad \text{given}$$

$$\boldsymbol{u}^1 = \boldsymbol{u}^0 + h_t \boldsymbol{w}_0 + \frac{(ch_t)^2}{2}\left[-A\boldsymbol{u}^0 + \boldsymbol{f}^0\right]$$

$$\boldsymbol{u}^{k+1} = 2\boldsymbol{u}^k - \boldsymbol{u}^{k-1} + (ch_t)^2\left[-A\boldsymbol{u}^k + \boldsymbol{f}^k\right]$$

Convergence of the three-point time-stepping scheme

- Time-stepping interations converge to the exact solution on $t \in [0, T]$ as $h_t \to 0$

Stability of time-stepping for wave equation with FDM Laplacian

Time-stepping with FDM Laplacian

We consider the iterations

$$\boldsymbol{u}^{k+1} = 2\boldsymbol{u}^k - \boldsymbol{u}^{k-1} - (ch_t)^2 A\boldsymbol{u}^k, \quad \boldsymbol{u}^k \in \mathbb{R}^n$$

Assuming $A = V\Lambda V^{-1}$ exists, let $\boldsymbol{u}^k = V\boldsymbol{b}^k$, same as in the method of lines. Then, we obtain iterations for $\boldsymbol{b}^k$:

$$\boldsymbol{b}^{k+1} = 2\boldsymbol{b}^k - \boldsymbol{b}^{k-1} - (ch_t)^2 \Lambda\boldsymbol{b}^k, \quad \boldsymbol{b}^k \in \mathbb{R}^n$$
$$b_j^{k+1} = 2b_j^k - b_j^{k-1} - (ch_t)^2 \lambda_j b_j^k, \quad j = 1, \ldots, n$$

Let $b_j^k = (b_j)^k$, i.e., b_j to the power k. Dividing by $(b_j)^{k-1}$ and solving the quadratic equation for b_j:

$$(b_j)^2 - \left[2 - (ch_t)^2\lambda_j\right] b_j + 1 = 0$$
$$b_j = \frac{1}{2}\left[2 - (ch_t)^2\lambda_j \pm \sqrt{(2 - (ch_t)^2\lambda_j)^2 - 4}\right]$$

Condition on exact b_j

- In the exact method-of-lines solution $|b_j(t_k)|$ are oscillating in time. There is no exponential decay/growth.
- Hence, in the numerical time stepping we need to have $|b_j| = 1$, so that $|(b_j)^k| = |b_j|^k = 1^k = 1$, since otherwise we get exponential decay/growth of $(b_j)^k$ with k
- This is only possible if in the formula for b_j we set:

$$(ch_t)^2 \lambda_j = 4$$

Numerical dispersion

Hence, ideally, we would like to have

$$\frac{(ch_t)^2}{4}\lambda_j = 1$$

However, e.g., in the 1D case, we have

$$\lambda_j = \frac{4}{h_x^2}\sin^2\left(\frac{\pi j}{2N}\right),$$

and it is not possible to choose h_t and h_x in such a way that

$$\left(\frac{ch_t}{h_x}\right)^2 \sin^2\left(\frac{\pi j}{2N}\right) = 1, \quad \text{for all } \; j = 1, \ldots, n;$$

- Therefore, the numerical time-evolution coefficients b_j^k of different eigenvectors $\boldsymbol{v}_j$ will never be perfectly "synchronized" – *numerical dispersion*

CFL stability condition

At least we can avoid divergence over time by requiring:

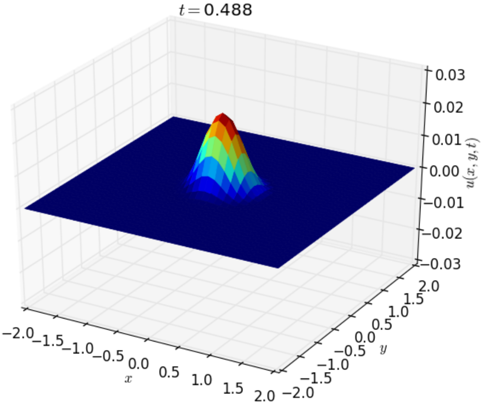
$$|b_j| \leq 1$$
$$\frac{(ch_t)^2}{4}\lambda_j \leq 1$$

This is a general condition. Example, in 1D case:

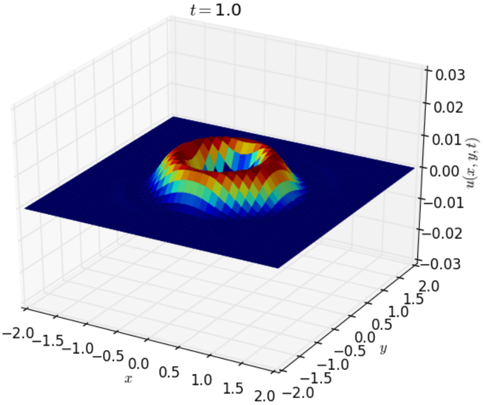
$$\frac{(ch_t)^2}{4}\frac{4}{h_x^2}\sin^2\left(\frac{\pi j}{2N}\right) \leq 1$$
$$\frac{(ch_t)^2}{h_x^2}\sin^2\left(\frac{\pi j}{2N}\right) \leq 1$$
$$\frac{(ch_t)^2}{h_x^2} \leq 1$$
$$0 \leq \frac{ch_t}{h_x} \leq 1$$

This is the *Courant-Friedrichs-Lewy* (CFL) stability condition in $1D$, and ch_t/h_x is called the *Courant number*.

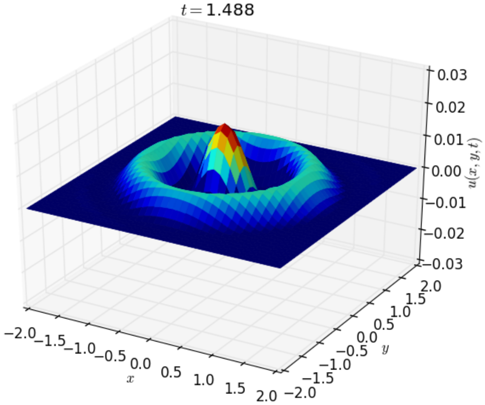
Wave motion: $t = 0.5$



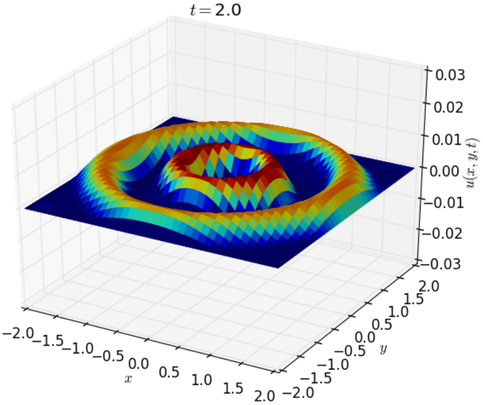
Wave motion: $t = 1.0$



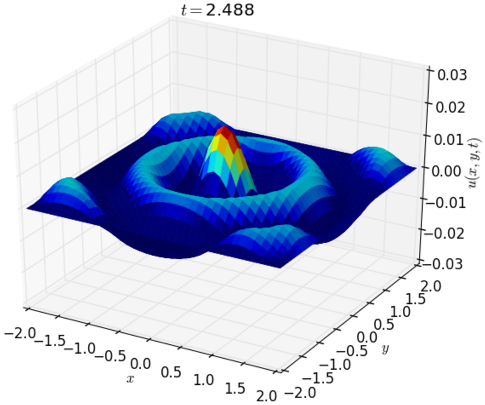
Wave motion: $t = 1.5$



Wave motion: $t = 2.0$



Wave motion: $t = 2.5$



Wave motion: $t = 3.0$